

PRINT DRAWER • RASTERIZE ONCE

# Flip the paper. Skip the wait.

The Print drawer now rasterizes each slide **once** and caches the images. Changing paper or orientation re-replaces those images onto the new sheet — no re-rasterize.

WHY IT WAS SLOW

# A paper change used to rebuild the whole PDF.

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- Every Letter → Legal → A4 flip re-ran the expensive step: **clone, embed fonts, rasterize** each slide.
- On a chart-heavy deck that is the dominant cost — paid again on every toggle.
- The pixels never changed, though — only where they land on the page did.

# Rasterize and assemble are now two steps.

## **Rasterize**

Each slide becomes a self-contained image at its native box, cached by render — not by paper.

## **Assemble**

Place the cached images on the chosen sheet: pure geometry, fit and centered.

WHAT THE FLIP COSTS NOW

# Re-place is a fraction of a rebuild.

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1×

RASTERIZE PER  
DECK

0

RE-RASTERIZES PER  
FLIP

3

PAPER SIZES REUSE  
IT

2

ORIENTATIONS REUSE  
IT

# The cache key is the render, not the sheet.

## **Paper size**

Re-replaces from cache — Letter, Legal, and A4 reuse the same images.

## **Orientation**

Re-replaces from cache — landscape and portrait share the pixels.

## **Color mode**

Re-renders — black & white is new ink, so the cache drops.

## **One tap**

Desktop still prints the vector deck in a single click.

ONE SOURCE OF TRUTH

# The drawer and the CLI assemble the same way.

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- `rasterizeDeckImages` and `assembleSheetPdf` live in the shared print kernel.
- The Print drawer caches images across flips; a Node export calls the same halves.
- `fitSlideOnSheet` places every slide — so the two surfaces can never disagree.

PRINT · SHARED KERNEL

# Configure freely. Build once.

*Rasterize once, assemble as often as you like — the  
boardroom handout is one tap away, on any paper.*